

Mapping Political articles: LDA and Social Network Analysis of Marcos' First Six Months (2025)

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I. INTRODUCTION

A. Background

The first six months of 2025 marked a politically turbulent and highly scrutinized period for President Ferdinand "Bongbong" Marcos Jr.'s administration. The country entered a year of midterm elections and national politics that revealed shifting alliances, executive-legislative tensions, and evolving public debates on accountability, funds, and governance. Major news outlets served as a central source of information when these topics unfolded, shaping the public's understanding of the administration's effect on people and the broader political movements across the country. However, it is imperative to note that the media can also limit the information with which audiences understand these issues and show dominant themes that can root from biases and limited outlet resources [1]. Traditional political analysis often focuses on individuals, groups, events, statements, and large amounts of information. As time passed, the information became more complex, large, and necessary for public distribution. Therefore, computational methods such as topic modeling and social network analysis have become increasingly important in identifying themes, influential actors, and relationships within social media discourse.

This study maps the political news coverage of President Ferdinand Marcos Jr.'s term from January to June 2025 by combining Latent Dirichlet Allocation (LDA) and Social Network Analysis (SNA). LDA reveals dominant topics and shifts in media attention over time, while SNA uncovers dominant actors such as politicians and organizations through different points of analyses in the methods. By integrating these two approaches, the study aims to provide a systematic and data-driven understanding of how the Marcos administration moved and participated in the narrative in Philippines' diverse political landscape. Providing a foundation for the future analysis of the emerging and disruptive politics is also one major goal of this study.

Media analysis of Philippine political discourse has increasingly relied on computational methods like Latent Dirichlet Allocation to uncover themes of their research approaches. Studies applying LDA to presidential speeches have revealed dominant topics in economic development and public service. In President Ferdinand Marcos Jr.'s presidency, news high-

lighted Marcos-Duterte tensions even before 2025 [2]. Recent events also proven the call for an in-depth analysis of the political landscape of the country while focusing on these two strongly influential families.

B. Research Questions

- 1) What dominant topics emerge in Philippine news coverage of President Marcos during his first six months of 2025?
- 2) Which actors (politicians, agencies, organizations) appear most central in the political discourse network, and how does their influence evolve over the six-month period?
- 3) How does Philippine news coverage of President Marcos change over the first six months of 2025, as revealed through temporal LDA modeling and social network analysis?

II. METHODOLOGY

A. Dataset Description

The dataset includes news articles from Philippine Star and Rappler, published between January 2025 to June 2025, and categorized under the topic of President Ferdinand Marcos Jr. These articles were collected through web scraping using BeautifulSoup, a Python library for parsing HTML and extracting data from web pages. The data was then cleaned to remove stopwords, punctuation, special characters, and other irrelevant elements, ensuring that the text was suitable for analysis. It was then organized per month for analysis.

B. Analysis

The gathered articles were analyzed using both Latent Dirichlet Analysis (LDA) for topic modeling, and AutoMap and ORA for social network analysis.

For topic modeling, the text data was vectorized using CountVectorizer and used to train an LDA model with a specified number of topics, producing both topic-term and document-topic distributions. From these outputs, the dominant topics for each month was identified. The results were then visualized using pyLDAvis, a visualization library for LDA under Python, allowing for a clearer understanding of the themes found within the dataset.

For Social Network Analysis, the text data was fed to AutoMap's Model Wizard for the initial generation of the thesaurus. After, it was ran again to include the generated thesaurus and stopwords (English and Filipino) for General and Project-Specific Refinements. The result of the Model Wizard, the XML files, were fed to ORA-LITE. Finally, the reports were generated for the meta and semantic network

III. RESULTS & DISCUSSION

A. January - Topic Modeling

For the month of January, the following topics were discovered:

- 1) Budget Politics and Marcos–Duterte Tensions in Government Administration
- 2) Philippine Foreign Policy and Security Relations under the Marcos Administration
- 3) Executive Clemency and Political Rehabilitation in the Marcos Administration

For the first topic, Budget Politics and Marcos–Duterte Tensions in Government Administration, the identified words are *duterte, say, budget, philippine, national, government, claim, 2024, make, vice, year, administration, iyong, also, blank, fund, house, first, 2025, dutertes, former, case, jr, project, drug, video, right, political, would, need.*

This topic explores the growing tension between the Marcos and Duterte camps, particularly surrounding government budget issues. Additionally, broader issues on budget transparency, allocation priorities, and administrative accountability are highlighted.

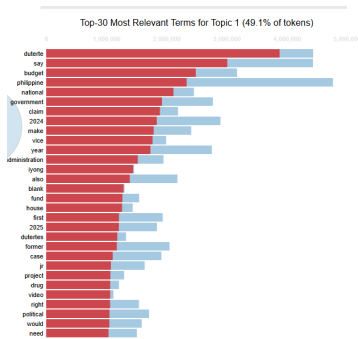


Fig. 1. Most Relevant Terms of Topic 1 under January

For the second topic, Philippine Foreign Policy and Security Relations under the Marcos Administration, the identified words are *philippine, trump, inc, january, say, country, china, first, year, continue, one, system, political, also, donald, missile, late, make, new, jr, state, manila, security, unite, 2024, former, member, work, rally, would.*

This topic explores how President Marcos Jr. has redefined the country's foreign policy and security posture. It also examines how these foreign policy moves impact regional security, economic interests, and the Philippines' position in global diplomacy.

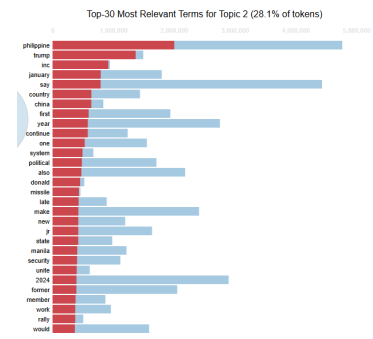


Fig. 2. Most Relevant Terms of Topic 2 under January

For the third topic, Executive Clemency and Political Rehabilitation in the Marcos Administration, the identified words are *mabilog, clemency, budget, 2024, executive, say, grant, government, case, court, city, former, administrative, iloilo, office, year, philippine, 2025, duterte, growth, ombudsman, country, go, mabilogs, also, tow, supreme, criminal, mayor, order.*

This topic especially focuses on Marcos' usage of executive clemency for former Iloilo City Mayor Jed Mabilog, a person targeted by the Dutertes during their anti-drug campaign. This case raises questions about the possible political motives and the behind this decision.

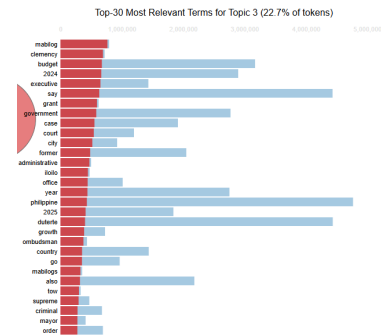


Fig. 3. Most Relevant Terms of Topic 3 under January

B. February - Topic Modeling

For the month of February, the following topics were discovered:

- 1) Marcos' Attitude Towards the Impeachment Trials of Vice President Sara Duterte
- 2) Campaign Activities and Political Messaging
- 3) Commemoration of the EDSA People Power Revolution

For the first topic, Marcos' Attitude Towards the Impeachment Trials of Vice President Sara Duterte, the identified words are *impeachment, say, house, duterte, complaint, philippine, one, representative, also, february, senate, first, vice, file, dutertes, public, sign, chief, lawmaker, sara, speaker, come, case, late, office, impeach, would, statement, make, allege.*

This topic is all about the discussions on the possible Impeachment of Vice President Sara Duterte. Coverage highlights how Marcos was continuously avoiding and distancing himself from commenting at talks of impeachment.

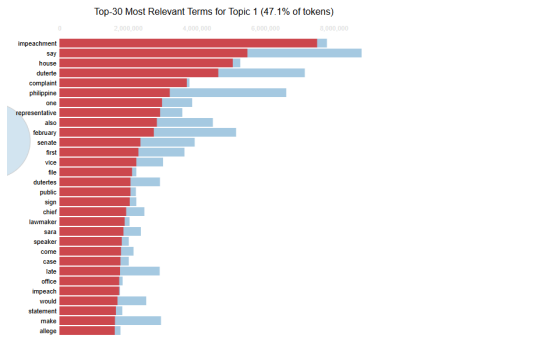


Fig. 4. Most Relevant Terms of Topic 1 under February

For the second topic, Campaign Activities and Political Messaging, the identified words are *campaign*, *candidate*, *philippine*, *say*, *senator*, *city*, *alyansa*, *po*, *duterte*, *senatorial*, *davao*, *administration*, *jr*, *voter*, *pilipinas*, *february*, *norte*, *vote*, *bagong*, *senate*, *slate*, *former*, *ilocos*, *province*, *also*, *region*, *bet*, *manila*, *coalition*.

This topic focuses on the starting campaign activities for the 2025 Midterm Election. Coverage includes rallies and other campaigning activities in areas such as Negros Occidental, Davao, Ilo-Ilo, and Ilocos Norte.

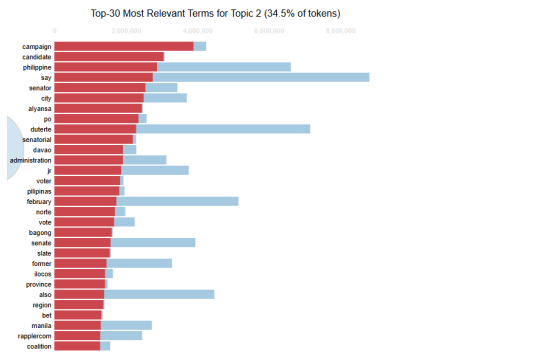


Fig. 5. Most Relevant Terms of Topic 2 under February

For the third topic, Commemoration of the EDSA People Power Revolution, the identified words are *rice*, *people*, *edsa*, *power*, *february*, *say*, *school*, *price*, *philippine*, *day*, *revolution*, *declare*, *government*, *emergency*, *project*, *ayuda*, *administration*, *food*, *special*, *program*, *25*, *jr*, *usaid*, *2024*, *secretary*, *history*, *1986*, *time*, *work*, *take*.

This topic explores the observance of the EDSA People Power Revolution and its significance as an event under the Marcos Administration. Additionally, coverage also highlights criticisms from various sectors that condone Marcos' attempts to downplay the revolution.

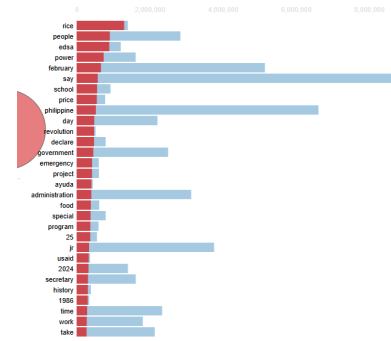


Fig. 6. Most Relevant Terms of Topic 3 under February

C. March - Topic Modeling

For the month of March, the following topics were discovered:

- 1) Duterte ICC Arrest and Political Fallout
- 2) BARMN Leadership Transition and Peace Governance
- 3) Infrastructure and Education Challenges Under Marcos

For the first topic, Duterte ICC Arrest and Political Fallout, the identified words are *duterte*, *senator*, *arrest*, *dutertes*, *say*, *philippine*, *icc*, *former*, *march*, *court*, *international*, *senate*, *administration*, *also*, *campaign*, *make*, *vice*, *one*, *first*, *would*, *rodrigo*, *government*, *alyansa*, *post*, *criminal*, *time*, *claim*, *drug*, *interpol*.

This topic is all about the developments surrounding the ICC Arrest of Former President Rodrigo Duterte and the possible political repercussions. Coverage highlights the different discussions on how the arrest of Duterte might affect Marcos' political positioning and relationship with other politicians.

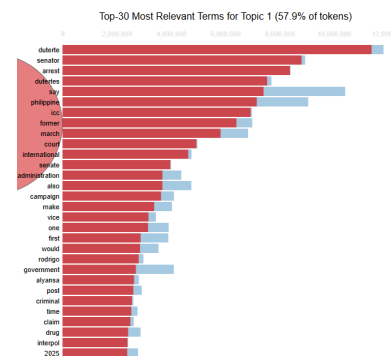


Fig. 7. Most Relevant Terms of Topic 1 under March

For the second topic, BARMN Leadership Transition and Peace Governance, the identified words are *bangsamoro*, *region*, *philippine*, *government*, *milf*, *chief*, *barmn*, *minister*, *macacua*, *first*, *also*, *jr*, *march*, *leadership*, *new*, *interim*, *lead*, *sur*, *member*, *peace*, *parliament*, *right*, *transition*, *appointment*, *development*, *dont*, *city*, *ebrahim*, *camarines*.

This topic focuses on the leadership reshuffle that happened in the Bangsamoro Autonomous Region in Muslim Mindanao.

Coverage especially highlighted Marcos' appointment of the new key officials that will help in running the region.

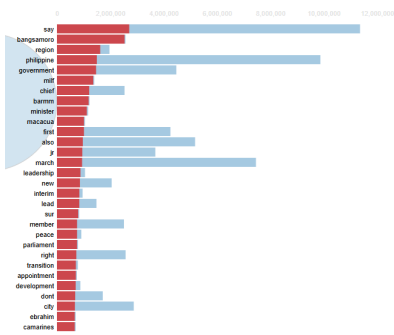


Fig. 8. Most Relevant Terms of Topic 2 under March

For the third topic, Infrastructure and Education Challenges Under Marcos, the identified words are *bridge*, *philippine*, *say*, *design*, *duterte*, *structural*, *student*, *project*, *suspension*, *collapse*, *administration*, *drug*, *jr*, *truck*, *call*, *flaw*, *work*, *country*, *also*, *support*, *university*, *one*, *march*, *year*, *education*, *overload*, *sea*, *use*, *start*, *dutertes*.

This topic explores the systemic challenges in infrastructure and education during the Marcos administration. Coverage include the collapse of the Cabagan-Sta. Maria Bridge and the need for more and improved classrooms in the country.

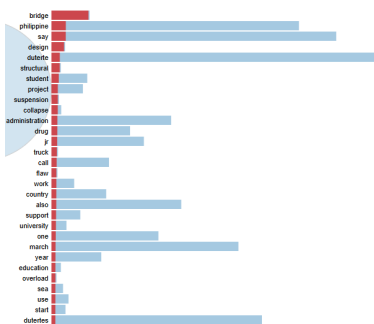


Fig. 9. Most Relevant Terms of Topic 3 under March

D. April - Topic Modeling

For the month of April, the following topics were discovered:

- 1) Duterte, ICC Arrest, and the 2025 Election Landscape
- 2) Marcos, State Visits, and Public Engagements
- 3) National Laws, Economic Development, and Governance Initiatives

For the first topic, Duterte, ICC Arrest, and the 2025 Election Landscape, the identified words are *duterte*, *former*, *mayor*, *city*, *philippine*, *trust*, *political*, *march*, *candidate*, *run*, *vice*, *senator*, *also*, *make*, *icc*, *election*, *campaign*, *cebu*, *national*, *arrest*, *say*, *romualdez*, *approval*, *point*, *survey*, *dutertes*, *sara*, *jr*, *rating*, *go*.

This topic focuses on Sara Duterte, former local leaders, and national political figures in the context of the 2025 elections. Coverage highlights Duterte's ICC-related legal challenges, arrest proceedings, and public statements, as well as campaign activities in Cebu and other key provinces.

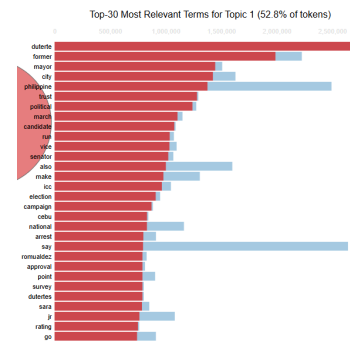


Fig. 10. Most Relevant Terms of Topic 1 under April

For the second topic, Marcos, State Visits, and Public Engagements, the identified words are *duterte*, *former*, *mayor*, *city*, *philippine*, *trust*, *political*, *march*, *candidate*, *run*, *vice*, *senator*, *also*, *make*, *icc*, *election*, *campaign*, *cebu*, *national*, *arrest*, *say*, *romualdez*, *approval*, *point*, *survey*, *dutertes*, *sara*, *jr*, *rating*, *go*.

This topic explores President Marcos Jr.'s interactions with international figures and the Filipino public in April 2025. Coverage includes legislative actions and outreach programs, highlighting the administration's efforts to connect with citizens

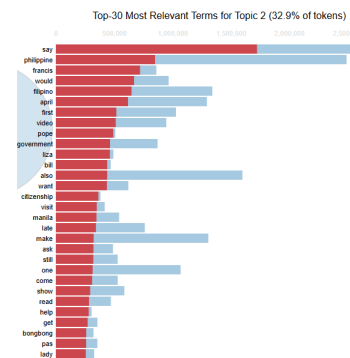


Fig. 11. Most Relevant Terms of Topic 2 under April

For the third topic, National Laws, Economic Development, and Governance Initiatives, the identified words are *philippine*, *national*, *law*, *also*, *april*, *department*, *say*, *plan*, *manila*, *sign*, *legal*, *case*, *economic*, *pas*, *order*, *free*, *ishiba*, *act*, *new*, *development*, *jr*, *force*, *public*, *official*, *conduct*, *assistance*, *minister*, *country*, *establish*.

This topic focuses on legal, economic, and administrative measures undertaken by the Marcos administration in April 2025. Coverage includes the signing of new laws, creation of

sara, serve, go, make, philippine, would, ally, aquino, house,
year, survey, first, know, vote, drug.

Term	Approximate Tokens
philippine	2,600
rational	1,800
law	1,500
also	1,400
port	1,300
department	400
may	300
gale	200
manila	800
sign	700
legal	600
case	500
economic	400
pas	300
order	200
tax	100
infila	100
act	100
new	100
development	100
y	1,500
time	100
public	100
office	800
conduct	100
assistance	100
member	100
country	500
station	100
rapplacon	1,000

Top 30 Most Relevant Terms for Topic 2 (36.3% of tokens)

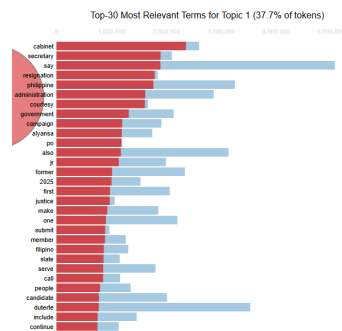
Term	Frequency (approx.)
say	4,800
debate	2,200
implementation	4,200
also	2,800
election	2,500
police	1,200
administration	2,800
public	1,200
that	1,200
vice	1,200
middle	1,200
series	1,200
former	1,200
candidate	2,200
term	1,200
serve	1,200
go	1,200
make	1,200
philippines	2,800
world	1,200
city	1,200
aquino	1,200
house	1,200
year	1,200
survey	2,200
last	1,200
know	1,200
role	1,200
drug	1,200

E. May - Topic Modeling

- 1) Cabinet Reshuffle and Political Realignment After Midterms
- 2) Impeachment Talk and Political Accountability After Elections
- 3) Provincial Power and Electoral Alliances

For the third topic, Provincial Power and Electoral Alliances, the identified words are *cebu, city, say, duterte, philippine, one, vote, governor, garcia, government, first, district, also, former, get, mayor, laoag, claim, province, candidate, law, public, icc, vice, include, team, jr, gold, provincial, ally*.

This topic examines the cabinet reshuffle and political adjustments that happened after the 2025 Midterm Elections. Coverage includes courtesy resignations from Cabinet officials and the replacement of important positions such as the PNP Chief and OWWA head.



For the second topic, Impeachment Talk and Political Accountability After Elections, the identified words are *say, duterte, impeachment, also, election, police, pnp, administration, political, chief, vice, midterm, senate, former, candidate*.

F. June - Topic Modeling

For the month of June, the following topics were discovered:

- 1) Philippine Government Initiatives and Public Service in June
- 2) Raul Villanueva and Supreme Court Appointments
- 3) Marcos' Governance, Policies, and Political Oversight

For the first topic, Philippine Government Initiatives and Public Service in June, the identified words are *philippine*,

also, government, june, new, visit, service, first, po, school, one, area, country, duterte, filipino, manila, would, freedom, jr, medium, public, report, city, right, see, national, secretary, could, administration .

This topic highlights the Philippine government's activities and public service initiatives under the Marcos administration during June 2025. Coverage includes Marcos' visits to schools and communities in Manila and other areas, efforts to improve national services and infrastructure, and partnerships with local and international agencies.

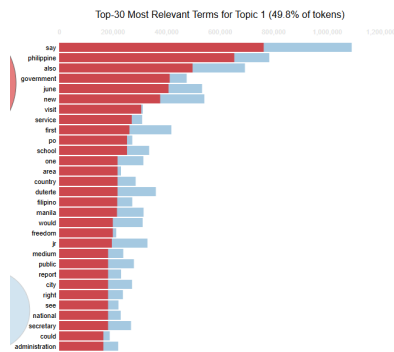


Fig. 16. Most Relevant Terms of Topic 1 under June

For the second topic, Raul Villanueva and Supreme Court Appointments, the identified words are *justice, court, villanueva, associate, bill, say, high, also, free, law, appointment, sc, first, administrator, new, survey, college, serve, duterte, judge, would, chief, case, appoint, senate, education, former, state, honor, university*.

This topic focuses on Marcos' judicial appointments, particularly the appointment of Raul Villanueva as Associate Justice and Court Administrator, marking his first Supreme Court pick. Coverage highlights Villanueva's long career in the judiciary, his potential impact on the legal field.

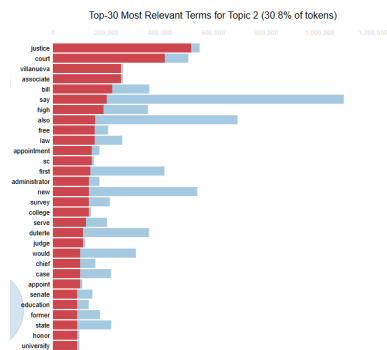


Fig. 17. Most Relevant Terms of Topic 2 under June

For the third topic, Marcos' Governance, Policies, and Political Oversight, the identified words are *say, power, secretary, june, cabinet, case, courtesy, resignation, philippine, week, tell, manila, jr, investigation, order, bersamin, last, give,*

presidential, continue, make, national, committee, executive, police, si, take, also, siquijor.

This topic covers key political developments and government actions under President Marcos in 2025, including executive decisions, cabinet and judiciary appointments, legislative updates, and public accountability issues.

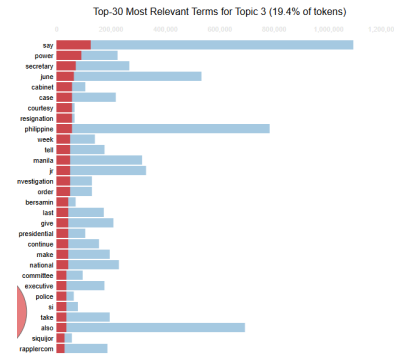


Fig. 18. Most Relevant Terms of Topic 3 under June

G. January - SNA

Liza Marcos led betweenness centrality as top agent connector, with Ernesto Torres as a key ally; authority metrics highlighted her bridging role. Semantic analysis tied top concepts to leadership, governance, religion, and church, framing early tensions around budget and clemency.

Rank [▲]	Agent	Value	Unscaled
1	liza_marcos	0.154	1,544.145
2	house_report_dwight_de_leon	0.119	1,193.324
3	executive_director_ernesto_torres_jr	0.104	1,037.935
4	us_president-elect_donald_trump	0.091	909.587
5	philippine_constitution	0.091	907.232

Fig. 19. Betweenness Centrality

H. February - SNA

Agent-event networks showed high out-degree/hub centrality for specialaction, frenchrevolution, and post-martiallaw agents, driving impeachment and EDSA narratives. Phrases acted as narrative drivers, with consistent Marcos-Duterte criticism (e.g., "pangulong ferdinand marcos jr and vice president sara dutertes impeachment").



Fig. 20. Actor x Actor

[illegible]

Fig. 21. Communicative Power Analysis

I. March - SNA

Authority-centrality surged for Sara Duterte/Dutertes in agent-agent ties, with heightened evokability to ICC/Hague following Rodrigo Duterte's arrest; US President Donald Trump appeared as a high-connection outlier influencing security discourse. Agent-unknown and Bangsamoro-frequency networks reflected BARMM transition uncertainty and infrastructure fallout (e.g., bridge collapse), amplifying political repercussions on Marcos' positioning. Polarization deepened, tying local crises to international legal pressures.

#	Feature	Coverage	Stability	Stability	Response	Weak Correlation Change
1	age	0.408	0.278	0.009	833.707/0.2	1
2	sex	0.501	0.235	0.040	87.0	207
3	sex_predict_gender	0.501	0.235	0.040	87.0	207
4	name	0.501	0.235	0.040	87.0	207
5	name	0.501	0.235	0.040	87.0	207
6	name	0.501	0.235	0.040	87.0	207
7	name	0.501	0.235	0.040	87.0	207
8	name	0.501	0.235	0.040	87.0	207
9	name	0.501	0.235	0.040	87.0	207
10	name	0.501	0.235	0.040	87.0	207
11	name	0.501	0.235	0.040	87.0	207
12	name	0.501	0.235	0.040	87.0	207
13	name	0.501	0.235	0.040	87.0	207
14	name	0.501	0.235	0.040	87.0	207
15	name	0.501	0.235	0.040	87.0	207
16	name	0.501	0.235	0.040	87.0	207
17	name	0.501	0.235	0.040	87.0	207
18	name	0.501	0.235	0.040	87.0	207
19	name	0.501	0.235	0.040	87.0	207
20	name	0.501	0.235	0.040	87.0	207
21	name	0.501	0.235	0.040	87.0	207
22	name	0.501	0.235	0.040	87.0	207
23	name	0.501	0.235	0.040	87.0	207
24	name	0.501	0.235	0.040	87.0	207
25	name	0.501	0.235	0.040	87.0	207
26	name	0.501	0.235	0.040	87.0	207
27	name	0.501	0.235	0.040	87.0	207
28	name	0.501	0.235	0.040	87.0	207
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37	name	0.501	0.235	0.040	87.0	207
38	name	0.501	0.235	0.040	87.0	207
39	name	0.501	0.235	0.040	87.0	207
40	name	0.501	0.235	0.040	87.0	207
41	name	0.501	0.235	0.040	87.0	207
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48	name	0.501	0.235	0.040	87.0	207
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50	name	0.501	0.235	0.040	87.0	207
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55	name	0.501	0.235	0.040	87.0	207
56	name	0.501	0.235	0.040	87.0	207
57	name	0.501	0.235	0.040	87.0	207
58	name	0.501	0.235	0.040	87.0	207
59	name	0.501	0.235	0.040	87.0	207
60	name	0.501	0.235	0.040	87.0	207
61	name	0.501	0.235	0.040	87.0	207
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64	name	0.501	0.235	0.040	87.0	207
65	name	0.501	0.235	0.040	87.0	207
66	name	0.501	0.235	0.040	87.0	207
67	name	0.501	0.235	0.040	87.0	207
68	name	0.501	0.235	0.040	87.0	207
69	name	0.501	0.235	0.040	87.0	207
70	name	0.501	0.235	0.040	87.0	207
71	name	0.501	0.235	0.040	87.0	207
72	name	0.501	0.235	0.040	87.0	207
73	name	0.501	0.235	0.040	87.0	207
74	name	0.501	0.235	0.040	87.0	207
75	name	0.501	0.235	0.040	87.0	207
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192	name					

Fig. 22. Communicative Power Analysis

Rank [▲]	Agent	Value	Unscaled
1	bongbong_marcos	0.652	0.461
2	sebastian_duterte	0.568	0.402
3	senator	0.441	0.312
4	philippines	0.411	0.290
5	ice_warrant_jr	0.338	0.239
19	vice_president_sara_duterte		0.106

Fig. 23. Agent x Agent

J. April - SNA

Hub centrality in agent-location networks stressed information dissemination, connecting President Bongbong Marcos to businessman Tony Tan Caktiong; Sara Duterte topped semantic ranks consistently amid election landscapes and ICC proceedings. Concepts like "best" and "bills" linked Marcos/VP to state visits and new departments, with networks showing sustained Duterte influence despite impeachment threats. Discourse hubs emphasized Cebu/provincial spreads, testing midterm loyalties.

Rank	Concept	Connections	Uncovered	Exhaustible	Uncovered	Frequency	Uncovered	Most Connected Concepts
1	houghling_martin	0.643	1.720	0.081	300.803	125	1	advertising (1), advertisements (1), advertisment (1), advertis (1), admes (1)
2	polytechnic_univ_geneva	0.366	1.450	0.051	366.764	50	0.774	advertisement (1), advertising (1), advertisements (1), advertisment (1), advertis (1), admes (1)
3	univ_geneva_universite_rattazylizy	0.289	839	0.017	128.473	0.474	92	ad (1), advertising (1), advertisements (1), advertisment (1), advertis (1), admes (1)
4	geneva_univ	0.173	630	0.011	85.651	0.838	83	advertising (1), advertisements (1), advertisment (1), advertis (1), admes (1)
5	fordham_univ_ny	0.138	548	0.006	45.078	0.416	79	advertising (1), advertisements (1), advertisment (1), advertis (1), admes (1)

Fig. 24. Communicative Power Analysis

K. May - SNA

Marcos-Duterte antagonism peaked in agent-agent networks, elevating allies; Ronald Dela Rosa led as top hub/curator in agent-organization ties (PNP, Senate), curating post-midterm narratives amid US political echoes (Trump/Vance). Task-task authority favored "serve, submit, lead" during cabinet reshuffles and provincial wins (e.g., Cebu governors), where impeachment prosecutors succeeded despite Sara's campaigns. Sara dropped to 4th in ranks, signaling temporary influence dip post-polls.

Rank	Agent	Value	Unscaled
1	chief-turned-senator_ronald_dela	0.506	0.358
2	bongbong_marcos	0.329	0.233
3	operations_police_lieutenant_general_robert_rodriguez	0.290	0.205
4	united_states_vice_president_id_vance	0.251	0.178
5	ferdinand_marcos_jr	0.152	0.108
6	media-averse_first	0.133	0.094
7	house_secretary_general_reginald_velasco	0.123	0.087
8	philippines	0.121	0.085
9	torre	0.111	0.079
10	director	0.099	0.070

Fig. 25. Agent x Organization

L. June - SNA

Agent-agent networks integrated newcomers like Chief Justice Gesmundo, Reginald Velasco (House Secretary), and Raul Villanueva (Supreme Court appointee), reducing volatility; Sara Duterte reclaimed top communicative power. Semantic analysis retained judicial/administrative clusters (oversight, investigations), framing governance continuity amid policy actions. Networks consolidated around accountability, with Duterte's persistence underscoring rivalry.

Rank	Concept	Connections	Usability	Exclusivity	Usual	Frequency	Used	Most Connected Concepts
1	via_cnn_model_state	0.360	1.087	0.000	258,228.09	1	116	activation (1), bias (3), relu (3), sigmoid (3)
2	avg	0.298	792	0.021	100,108.64	1	116	best (2), cnn (2), state (2), sigmoid (2), avg (1), mean (1)
3	tensorflow_session	0.214	967	0.047	228,642.29	1716	83	activation (3), best (1), bias (1), relu (1), sigmoid (1)
4	nn	0.206	632	0.009	443,710.39	819	90	cnn (2), sigmoid (1), avg (1), state (1), tensorflow_session (1), tensorflow_ops (1)
5	nn	0.207	617	0.010	467,338.10	970	82	activation (2), tensorflow_session (2), tensorflow_ops (2)
6	tf.nn.nn_ops	0.208	618	0.017	79,888.40	630	66	sls (1), relu (1), relu (1), activation (1), best (1)
7	softmax_session_p	0.171	360	0.003	28,839.30	0.66	54	best (1), tensorflow_ops (1), softmax (1), tensorflow_session_p (1), tensorflow_ops (1)
8	choc_lstm_generate	0.122	734	0.004	26,178.70	0.41	48	avg (1), activation (1), bias (1), best (1), sigmoid (1), softmax (1)
9	avg	0.124	474	0.005	23,029.92	0.52	42	best (1), decision (1), goal (1), happy (1), softmax (1), tensorflow_session_p (1)
10	hvae_session_gmm_reginal_video	0.142	478	0.009	44,907.64	0.284	33	best (1), tensorflow_session_p (1), tensorflow_session_p (1), tensorflow_session_p (1)

Fig. 26. Communicative Power Analysis

M. Evolution of Actor Influence - SNA

Core actors such as President Bongbong Marcos and Sara Duterte anchored networks, with Liza Marcos' early brokerage (Jan) yielding to Duterte surges (Feb-May, post-impeachment/arrest evokability via ICC/police). Influence trajectory: Marcos allies bridged fragmentation (Jan-Feb), Dutertes polarized midterms (Mar-May, hubs like Dela Rosa curating), then new officials (Gesmundo, Villanueva) stabilized June post-election. Organizations (Supreme Court, House, ICC) and internationals (Trump) gained centrality; Sara's rank fluctuations (top Apr/Jun, dip May) reflected resilient southern/OFW support despite attacks.

IV. CONCLUSION

A. Importance of Research

Mapping political articles from Marcos' first six months using LDA and Social Network Analysis is important for understanding how major news platforms collectively frame his presidency. LDA identifies dominant topics, revealing which recent issues receive the most coverage. Social Network Analysis shows how actors, organizations, and key terms are interconnected across the articles, highlighting the structure of media discourse. Using both methods together allows us to uncover patterns and relationships that might not be visible from reading individual articles. This approach provides a systematic, data-driven way to analyze media narratives and public perception of Marcos' presidency in the first half of 2025.

B. Answer to Research Questions

The results of the approaches revealed prominent and notable connections within the ' political landscape of the Philippines for the timeline studied. Here are the answers and questions for the research:

- 1) What dominant topics emerge in Philippine news coverage of President Marcos during his first six months of 2025?

The LDA revealed that news coverage across the six-period constantly centered on topics relating to executive-legislative conflict, national budget and fund controversies, local and national governance challenges, and public accountability issues. President Marcos Jr. was a central topic together with prominent figures such as Vice President Sara Duterte and the members of both legislative and executive branch.

In January, topics included Budget Politics and Marcos-Duterte Tensions (key terms: duterte, budget, government), Philippine Foreign Policy and Security Relations. February highlighted Marcos' stance on Sara Duterte's impeachment trials, 2025 midterm campaign activities and alliances, and the EDSA People Power commemorations. March covered Rodrigo Duterte's ICC arrest fallout and infrastructure/education challenges like construction and project implementation of government initiatives.

On the other half, April topics encompassed Duterte's ICC issues that are ties to the 2025 elections and Marcos engagements. Next, May focused on post-midterm cabinet reshuffles that revealed new political faces and resignation of some officials. From this, accountability debates formed and the people demanded transparency from officials. Lastly, June emphasized on government public services such as visitations and acknowledgement of concerns. Overarching themes aggregated as power struggles, party tensions, and government reform/election issues.

- 2) Which actors (politicians, agencies, organizations) appear most central in the political discourse network, and how does their influence evolve over the six-month period?

The SNA using AutoMap and ORA identified central actors in agent-agent, agent=event, and semantic networks in evolution tracked monthly. Liza Marcos topped betweenness-centrality in January as a key connector among its agents, alongside allies like Ernesto Torres. Following this semantic analysis are concepts connected to leadership, governance, religion, and large group movements. In February, agents like french revolution and post-martial law showed high out-degree and hub centrality in agent-event networks. This drove narratives and stronger tensions from the Marcos administration and historical accounts. March, on the other hand, highlighted increased evocability for Sara Duterte and Rodrigo Duterte.

An interesting agent was revealed in April as businessman Tony Tan Caktiong was connected to President Bongbong Marcos. However, Sara Duterte consistently dominated the topping lists as an important agent in location ties. Allies started to emerge within the Marcos vs. Duterte conflict with connections showing strong emergence from political families and high ranking executive and legislative officials. Ronald Dela Rosa became a top hub in these networks. June introduced Chief Justice Gesmundo and officials like Reginald Velasco; Sara Duterte regained top rank, with Raul Villanueva entering as a new judicial figure.

- 3) How does Philippine news coverage of President Marcos change over the first six months of 2025, as revealed through temporal LDA modeling and social network analysis?

Temporal LDA modeling detected progressive topic shifts, starting with institutional and policy-oriented coverage in January such as budget problems and numerous foreign policy realignments amid calls to delay pork barrel releases until after the midterms. By February, topics pivoted a different direction and towards political legitimacy challenges. The Marcos-Duterte conflict became clearer at this point where there was tension and avoidance of issues within the Marcoses and Sara Duterte's impeachment commentary. Additionally, EDSA commemorations was criticized for downplaying history, signaling rise in narrative tensions even from other political groups. March also reached a new crisis peak with Rodrigo Duterte's ICC arrest dominating discourse.

SNA completed the analysis by showing the network fragmentation and connections in January to February,

where Liza Marcos bridged agent gaps but semantic clusters revealed interesting connections about governance and religion ties amid Marcos-Duterte rifts. Polarization intensified in March to May, with Duterte centrality surviving post-arrest (evocability links to ICC/Hague), agent-event hubs driving election narratives, and May's agent-organization networks elevating Ronald Dela Rosa as an information curator during the tension and cabinet reshuffle. It is also notable that external influences from foreign policies are still present in these months in relation to US politics.

After all these events themes of stabilization occurred in June via LDA's focus on topics such as public services, judicial appointments and oversights that aligns with Marcos' vows for tangible changes and rising trust ratings. SNA reflected consolidation, introducing figures like Chief Justice Gesmundo while Sara Duterte retained top communicative power, this indicates the enduring rivalry but reduced volatility as new officials centralized networks. Overall, coverage evolved from reactive internal conflicts (Jan-Feb) through polarized crisis management (Mar-May, amid human rights critiques and economic shortfalls) to structured reform narratives (June), underscoring media framing of Marcos' presidency as adapting to elite pressures and institutional maneuvers. This dual lens reveals a trajectory toward post-midterm accountability, with persistent Duterte antagonism shaping public perception despite governance pivots.

C. Significant Insights

- 1) Media coverage reflects the tension between political conflict and governance. The persistent presence of topics related to internal political conflicts, especially involving the Marcos and Duterte camps, indicate that narratives that evolve within a specific period shapes connections and intent within it agents. This tension suggests that public attention during the first half of 2025 was heavily influenced by power struggles, which overshadowed policy discussion on both national and international levels.
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- 2) The political landscape became more fragmented leading to the midterm elections. In both LDA and SNA results indicate increasing complexity in political narratives as it evolved within the ins and outs of agents and their relationships. Whereas earlier coverage focused on executive disputes, later months saw a proliferation of actors and the issues they face as the narrative continues. This political environment calls for an analysis for its decentralized and competitive environment.
- 3) Central actors exerted disproportionate influence on discourse. SNA showed a small set of political elites and groups that consistently dominated the narrative space. As the main topic of this study rooting from President

Bongbong Marcos articles, it would not be discouraging to see him as a consistent agent. However, other actors that consistently dominated the results suggests that media coverage tends to reinforce existing centers of political power and stirrers of political narrative.

D. Future Work

The researches recommend the following for future work:

- 1) Extend the study to cover a full year once the year ends, or explore earlier periods.
- 2) Incorporate sentiment analysis to measure whether media coverage is positive, negative, or neutral.
- 3) Include additional news outlets, blogs, or social media platforms to create a more comprehensive map of political discourse and media influence.